



## Practical - 9 Count frequencies sorted

```

from pyspark.sql import SparkSession
from pyspark.sql.function import collect_set, col
from itertools import combinations

spark = SparkSession.builder \
    .appName("AmazonFood Frequent Pairs") \
    .getOrCreate()

data = spark.read.csv("Amazon-Food-Reviews.csv",
    header=True, inferSchema=True)

review = data.select("UserId", "ProductId").drop()

user_products = review.groupBy("UserId") \
    .agg(collect_set("ProductId").alias("products"))

def generate_pairs(products):
    return list(combinations(sorted(products),
        2))

pair_rdd = user_products.rdd.flatMap(lambda
    row: generate_pairs(row["products"]))

pairs_df = pair_rdd.map(lambda pair:
    (pair, 1)) \
    .toDF(["pair", "count"])
    
```



Pair-counts show (20, truncate = false)

Spurk stopc)